ONCE: Boosting Content-based Recommendation with Open- and Closed-source LLMs

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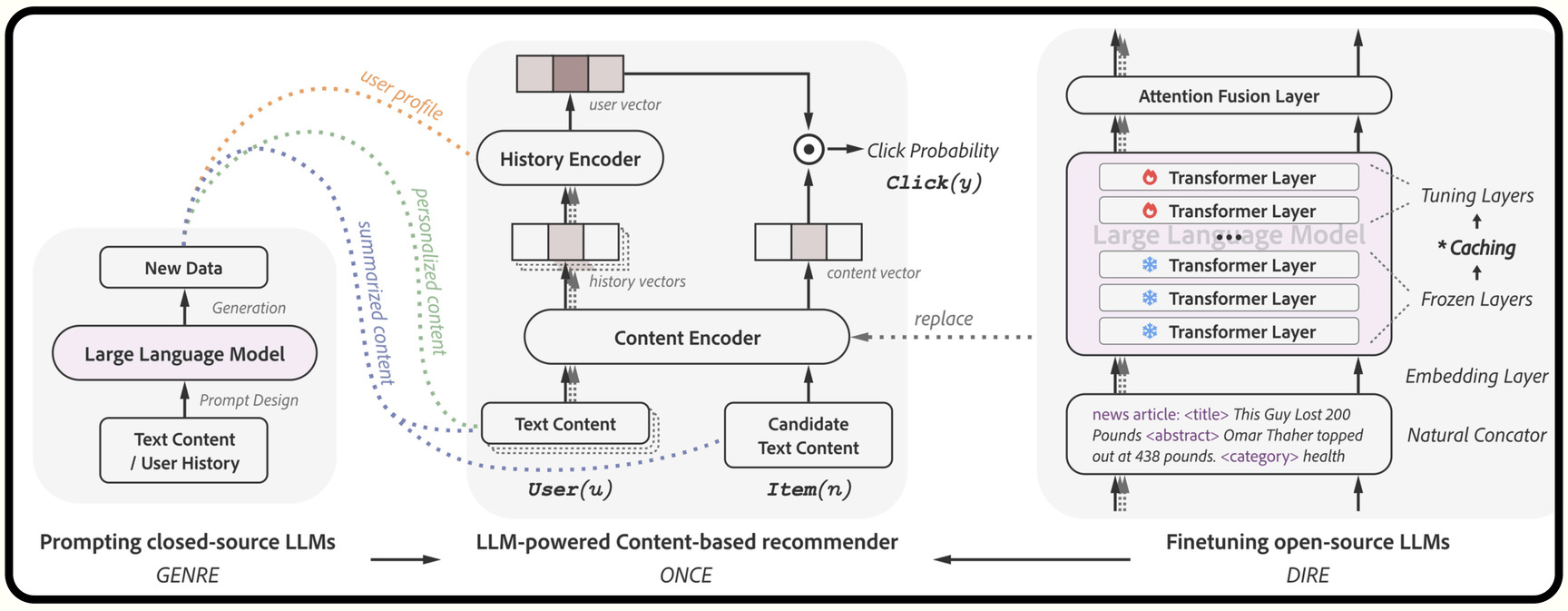


Fig. 1: Overview of ONCE framework, including open-source LLM to enhance recommendations (DIRE) and enriching prompts from closed-source LLMs (GENRE).

Dataset	EB-NeRD
	Original
# content	20,738
tokens/title	6.69
characters/title	42.0
# users	18,827
# new user	368
content/user	136
content/user$_n$	5.0
	Content Summarizer (CS)
tokens/title	+5,69
characters/title	+39,88
	User Profiler (UP)
topics/user	5.83
regions/user	1.07

Background

LLM-aided Recommender Systems

- Improved Personalization
- Contextual Understanding
- Scalability
- Analyzes unstructured data, to improve recommendation quality.

Background

EB-NeRD Dataset

- 2.7 million users, 600 million impression logs, over 120,000 news articles
- Components: User interaction logs, Click histories, Article features

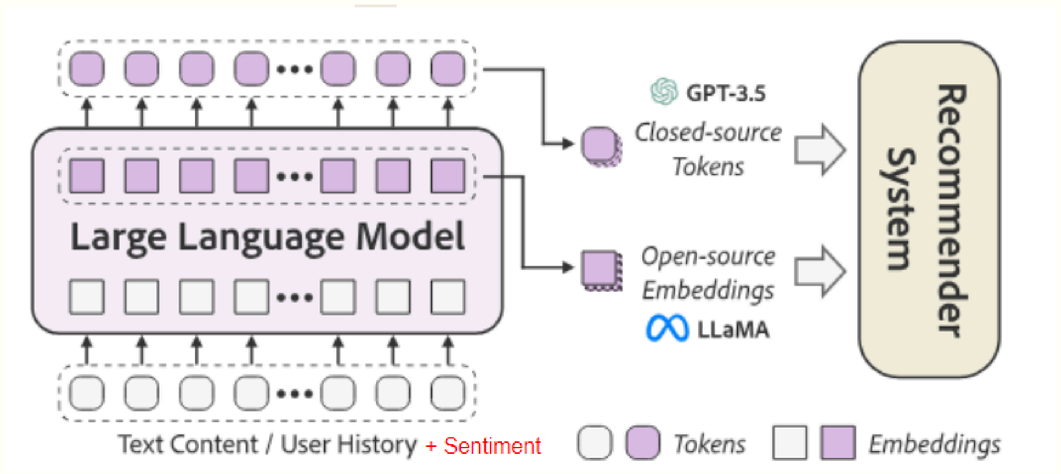
Architecture: DIRE and GENRE

- **DIRE:** Discriminative Recommendation Framework
  - Utilises open-source LLMs (LLaMA 7b) as content encoders
  - Enhances content representation at the embedding level
  - Key components: Embedding layer, Attention Fusion Layer, and Fine-tuning strategy
- **GENRE:** Generative Recommendation Framework
  - Uses closed-source LLMs (e.g., GPT-3.5) for data augmentation
  - Enhances training data at the token level
  - Key roles: Content Summarisers, User Profilers, Personalised Content Generators, Chain-based Generation

ONCE Intergration & Training

- ONCE Framework Overview
  - Combines strengths of both open- and closed-source LLMs
  - Synergistic approach to improve recommendation accuracy
- Training Strategy
  - Fine-tuning top layers of LLMs for computational efficiency
  - Use of Low-Rank Adaptation (LoRA) for efficient fine-tuning
  - Precomputed hidden states to reduce computational load

Extension: Sentiment Features



Replication

- **Baseline Models: NRMS and NAML (FastFormer)**
  - NRMS: Multi-head self-attention to capture word and news interactions
  - NAML: Attention-based model focusing on user preferences
  - FastFormer: Models global contexts via additive attention
- **Replication with DIRE and GENRE**
  - LLaMA 7B as content encoder in DIRE
  - Data augmentation with GPT-3.5 in GENRE
- **ONCE**

Table 2: Performance comparison of various models on the EB-NeRD dataset. (Part 1)

	Model	NRMS (2019c)				Fastformer (2021b)			
	Metric	AUC	MRR	N@5	N@10	AUC	MRR	N@5	N@10
	Original	-	-	-	-	-	-	-	-
DIRE	LLaMA7B (Ours)	55.77	34.55	38.53	46.30	55.95	34.75	38.72	46.47

Table 3: Performance comparison of various models on the EB-NeRD dataset. (Part 2)

	Model	NAML (2019a)			
	Metric	AUC	MRR	N@5	N@10
	Original	-	-	-	-
DIRE	LLaMA7B (Ours)	55.53	34.53	38.48	46.36

Conclusion and Future Work

- Leveraging both open and closed-source LLMs enhance recommendation systems performance
- ONCE framework's structure & robustness was able to be adjusted to various content-based domains such as the Eb-NerD dataset.
- **Limitations:** Computational cost of LLMs and implicit LLM biases (e.g., biased for US data) and privacy concerns (specifically for closed-source)
- **Future directions:** (1) Hybrid Approach - Integrate LLMs and traditional recommendation algorithms to enhance computational efficiency and mitigate model biases. (2) Real-time user-feedback to continuously personalise recommendations.

Liu, Q., Chen, N., Sakai, T., & Wu, X.-M. (2024). ONCE: Boosting Content-based Recommendation with Both Open- and Closed-source Large Language Models. In Proceedings of the Seventeenth ACM International Conference on Web Search and Data Mining.
Liu, Q.. (2023). Legommenders: A Modular Framework for Recommender Systems. Available at https://github.com/Jyonn/Legommenders